SIMATS ENGINEERING

ASSESSMENT TOOL 2 — ANSWERS

MLA0302 — Reinforcement Learning for Environment and Agent

CO2: Apply Dynamic Programming methods to solve RL problems and derive optimal policies

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Question 1

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. (5 Marks)

State Space

- Current GPS location of each idle driver
- Location and destination of each pending passenger request
- Local demand density (requests per zone) and time of day
- Traffic conditions / estimated travel time between zones

Action Space

- Assign a specific idle driver to a specific passenger request
- Reposition an idle driver toward a high-demand zone
- Hold/delay a match if a better pairing is likely soon

Reward Function

- + reward inversely proportional to passenger waiting time (shorter wait = higher reward)
- + small reward for high driver utilization (fewer idle drivers)
- − penalty for long detours or unfair driver-passenger distance mismatches
- − penalty for unmatched/cancelled requests

Explanation

By rewarding short wait times and penalizing idle/unmatched drivers, the agent learns to proactively position idle drivers near predicted demand hotspots instead of reacting only after a request arrives.

Over many episodes, the learned policy converges toward matches that jointly minimize passenger wait time and maximize driver utilization, which directly improves overall service quality.

Question 2

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. (5 Marks)

Exploitation

- Recommend material similar to what the learner has previously engaged with well (high completion rate, high score after study)
- Maximizes short-term relevance using known preferences

Exploration

- Occasionally recommend new topics, formats, or difficulty levels the learner hasn't tried
- Helps discover better-suited content and fills knowledge gaps the learner didn't know to ask for

Explanation

Pure exploitation traps the learner in a narrow loop of familiar content, causing stagnation and missed learning opportunities. Pure exploration wastes the learner's time on irrelevant material and reduces trust in the platform. An epsilon-greedy (or similar) strategy — mostly recommending known-effective material with a small, controlled probability of trying new content — balances reinforcing mastery with expanding the learner's knowledge base over time.

Question 3

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. (5 Marks)

States

- Current grid position / map coordinates
- Dirt level sensed at current location
- Battery level
- Distance to nearest obstacle

Actions

- Move forward/backward/left/right
- Activate suction / change cleaning mode
- Return to charging dock

Reward Function

- + reward for cleaning a dirty cell
- − penalty for collisions with obstacles
- − small penalty per time step (encourages efficiency)
- + reward for docking to recharge before battery depletes

Explanation

The policy is the mapping from each sensed state to the best action (e.g., 'if dirt detected ahead, move forward and clean; if obstacle detected, turn'). The value function estimates the total expected future reward from a given state, not just the immediate reward — so the agent learns to prefer paths that cover more uncleaned area overall, even if a nearby small reward looks tempting in the short term. As the agent explores the room repeatedly, value estimates are updated (e.g., via temporal-difference learning), and the policy derived from these estimates becomes increasingly efficient — covering the room faster with fewer repeated passes and fewer collisions.

Question 4

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. (5 Marks)

Key Reward-Design Challenges

- Sparse and delayed rewards — real profit/loss is only known once a position is closed, not at the moment of each action
- High market volatility and noise make it hard to distinguish a genuinely good decision from a lucky short-term price move
- Non-stationary environment — market behavior shifts over time, so a reward function tuned to past data may no longer reflect good decisions
- Risk is not captured by raw profit alone — a reward based only on price change ignores exposure/drawdown risk

Explanation

If the reward is defined purely as immediate price change after each action, the agent tends to over-trade on short-term noise, mistaking random fluctuations for exploitable patterns, and ignores portfolio risk. This can produce a policy that looks profitable in backtesting but takes oversized risky positions in practice, leading to large drawdowns or account 'blow-ups' when volatility spikes. Good reward design must incorporate risk-adjusted returns (e.g., penalizing large drawdowns or position sizes) rather than raw profit alone, to produce a stable, generalizable trading policy.

Question 5

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. (5 Marks)

State Space

- Current electricity demand per zone/region
- Available generation capacity (conventional + renewable)
- Energy storage / battery charge levels
- Time of day and weather forecast (affects solar/wind supply)

Action Space

- Increase/decrease supply allocation to a zone
- Charge or discharge storage/battery units
- Activate or deactivate backup generation
- Apply demand-response signals (e.g., dynamic pricing to shift load)

Reward Mechanism

- \+ reward for meeting demand reliably without shortages
- \+ reward for minimizing wasted/over-generated energy
- − large penalty for blackouts or load-shedding events
- − penalty for sharp peak-load spikes

Explanation

This reward mechanism directly balances the grid's two competing goals — reliability (always meeting demand) and efficiency (not wasting generated power) — while strongly discouraging blackouts, which guides the learned policy toward smooth, proactive load balancing rather than reactive emergency responses.

Question 6

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. (5 Marks)

States

- Patient vital signs (heart rate, blood pressure, blood glucose, etc.)
- Current medication dosage level
- Time since last dose
- Patient-specific history / response pattern

Actions

- Increase dosage
- Decrease dosage
- Maintain current dosage
- Alert a clinician for manual review

Rewards

- + reward for vitals remaining within the safe/target range
- − large penalty for an adverse reaction or vitals leaving the safe range
- − small penalty for unnecessary/frequent dosage changes (encourages stability)

Explanation

Because patients respond differently to the same dosage, a fixed one-size-fits-all schedule is often suboptimal. Through repeated interaction, the RL agent learns each patient's individual dose-response pattern and gradually adjusts dosing to keep vitals stable faster and with fewer adverse events than a static protocol — while the strong penalty on unsafe vitals keeps the learned policy conservative and safety-first.

Question 7

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. (5 Marks)

Exploitation

- Reuse routes that have previously proven fast and safe (known good paths)

Exploration

- Occasionally try alternative routes to discover shortcuts or adapt to new obstacles, weather, or no-fly zones

Explanation

If the drone only exploits known routes, it cannot adapt when conditions change (e.g., new construction blocking a previously optimal path), leading to persistently suboptimal or even unsafe navigation. If it explores too much, it wastes battery and delivery time testing unproven paths. A decaying exploration strategy — trying more alternatives early on, then increasingly favoring the best-known routes as confidence grows — lets the system discover efficient paths initially while still periodically re-checking for better options as the environment changes, which is essential for continuously optimized delivery routing.

Question 8

8. Design and develop a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. (5 Marks)

States

- Current assembly stage / step
- Part positions and orientation
- Robot arm position
- Sensor readings for misalignment/error

Actions

- Move arm to a target position
- Grip / release a part
- Insert or place a part
- Adjust applied force
- Verify assembly correctness

Reward Function

- + reward for each correctly completed assembly step
- − penalty for misalignment or assembly errors
- − small penalty per time step (encourages speed)
- + bonus for completing the full assembly within a target time

Explanation

The policy maps the robot's current sensed state to the best next action for the assembly task. The value function estimates the total expected future reward from a given state — not just the immediate step's reward — so the robot learns to favor actions that set up smoother, faster downstream steps (e.g., positioning a part in a way that eases the next insertion), rather than greedily optimizing only the current action. As value estimates are refined through repeated assembly cycles, the resulting policy becomes progressively faster and more error-free.

Question 9

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. (5 Marks)

Key Reward-Design Challenges

- Reward sparsity — clicks are rare relative to the number of ad impressions shown
- Delayed/attribution issues — a conversion (e.g., purchase) may happen long after the click, making it hard to credit the right ad/action
- Reward hacking risk — optimizing purely for clicks can favor sensational or misleading ('clickbait') ads
- Feedback-loop bias — the system only observes outcomes for ads it chose to show, biasing future learning

Explanation

If the reward function is based purely on click-through rate, the agent can learn to favor attention-grabbing but low-quality or misleading ads, which may raise short-term CTR while damaging user trust, brand reputation, and advertiser satisfaction in the long run. Reward sparsity also slows convergence, since useful learning signal (a click) is infrequent. This is typically mitigated with reward shaping — combining CTR with auxiliary signals like dwell time, conversion rate, or user satisfaction scores — to produce a policy that optimizes genuine engagement rather than exploiting the click-reward signal alone.

Question 10

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. (5 Marks)

States

- Current soil moisture level
- Weather forecast (rain probability, temperature)
- Crop growth stage
- Water reservoir/tank level

Actions

- Turn irrigation on/off for a zone
- Adjust water flow rate / duration

Reward Function

- + reward for maintaining soil moisture within the optimal range for the crop stage
- − penalty for water wastage/over-irrigation
- − penalty for under-watering (risk to crop health)
- + small bonus for water conservation when moisture is already adequate

Explanation

This reward function directly balances two competing goals — crop health (needs sufficient water) and resource conservation (avoiding waste) — so the learned policy irrigates only when needed and by the right amount, adapting to weather forecasts (e.g., skipping irrigation if rain is imminent) rather than following a fixed schedule.